



PHYSICS

Stage 2

WACE Examination 2012

Marking Key

Marking keys are an explicit statement about what the examiner expects of candidates when they respond to a question. They are essential to fair assessment because their proper construction underpins reliability and validity.

When examiners design an examination, they develop provisional marking keys that can be reviewed at a marking key ratification meeting and modified as necessary in the light of candidate responses.

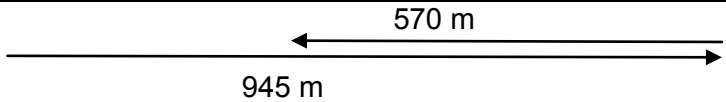
Section One: Short answers

40% (68 marks)

Question 1

(3 marks)

A hiker is walking east across a flat desert. After walking for 945 m, he realises he has dropped his water bottle and walks back 570 m to get it. Draw and label a vector diagram showing his path and calculate his resultant displacement.

Description	Marks
 Label 1 direction 1 $945 - 570 = 375 \text{ m}$	1–2
	1
Total	3

Question 2

(4 marks)

For the following pictures read the scale, record the measurement and determine the uncertainty of the measurement.

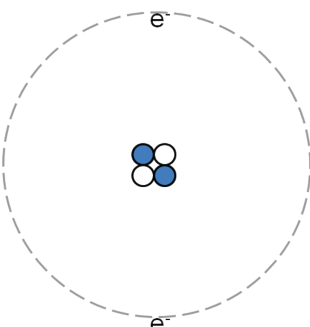
Description	Marks
0.9 (N)	1
0.05	1
124 (mL)	1
1	1
Total	4

Note: the term 'uncertainty' implies $\pm$ so magnitude only is required

Question 3

(3 marks)

Draw and label a diagram to show your understanding of the currently accepted structure of a helium - 4 atom.

Description	Marks
 sketch with:	1
All three labels = 2 marks; any two labels = 1 mark 2 electrons, labelled 2 protons, labelled 2 neutrons, labelled	1–2
Total	3

Question 4

(3 marks)

A sprinter completes one lap of an oval 400 m track in 55.0 s, finishing at the same point he started.

- (a) Determine the speed of the sprinter.

Description	Marks
Speed = distance/time = 400/55	1
= 7.27 m s ⁻¹	1
Total	2

- (b) Determine the velocity of the sprinter.

Description	Marks
v = s/t = 0/55 = 0 m s ⁻¹	1
Total	1

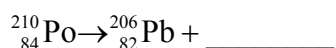
Question 5

(5 marks)

The radioactive isotope polonium-210 decays directly to lead-206, which is stable.

- (a) Complete the equation below showing this decay.

(2 marks)



Description	Marks
Identifies alpha decay particle as alpha, α , helium nucleus, He, etc.	1
Properly shows particle magnitudes as $\frac{4}{2}$	1
Total	2

- (b) The half-life of polonium-210 is 138 days. If a sample of polonium-210 has an initial activity of 2000 Bq, calculate how long will it take before the sample's activity decreases to 125 Bq.

(3 marks)

Description	Marks
Shows some calculations e.g. $A = A_0(\frac{1}{2})^n$ or 2000;1000;500;250;125 $125 = 2000(\frac{1}{2})^n$	1
Identifies number of half-lives = 4	1
$4 \times 138 = 552$ days	1
Total	3

Question 6

(4 marks)

A toaster malfunctions and does not eject the toasted bread. Describe the possible dangers associated with getting the toasted bread out of the toaster with a metal-handled knife.

Description	Marks
The toaster is still plugged in and may have a 'live' element in it	1
The metal knife will act as a contact and you will become conductor	1
A possible pathway for the current to flow is through you and into the ground	1
This could cause serious injuries (e.g. burns, cardiac arrest, etc)	1
Total	4

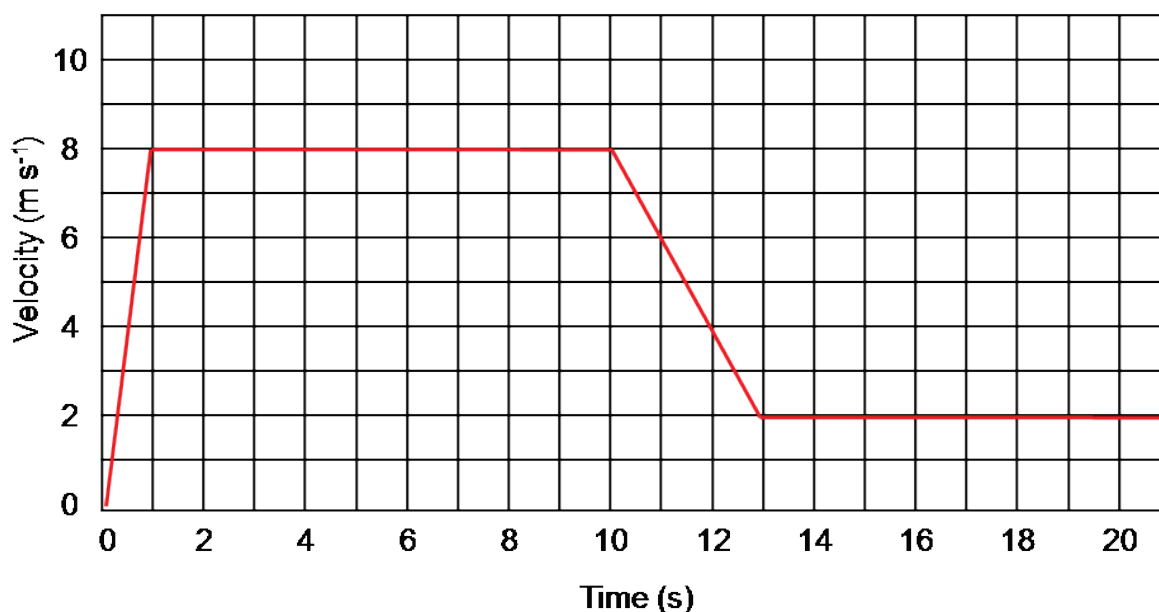
Note: other reasonable descriptions accepted

Question 7

(5 marks)

Jill begins from rest and after 1.0 s is moving at a velocity of 8.0 m s^{-1} for 9.0 s. She then takes 3.0 s to slow down to 2.0 m s^{-1} and continues at this velocity.

- (a) Graph Jill's motion on the velocity time graph below from the information given above. (2 marks)



Description	Marks
Times are appropriately spaced (x-coordinates)	1
magnitudes correct (y-coordinates)	1
Total	2

- (b) Calculate Jill's acceleration between the 10th and the 13th second. (3 marks)

Description	Marks
$a = (v-u)/t = (2.0-8.0)/3.0$	1
$a = -2.0 \text{ m s}^{-2}$	1
(direction e.g. opposite to motion or negative)	1
Total	3

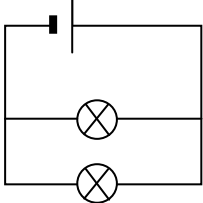
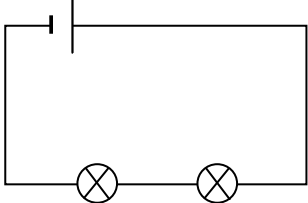
Question 8

(6 marks)

Complete **two (2)** circuit diagrams, which consist of one 1.5 V cell lighting two light globes.

- (a) Draw one circuit with the light globes in parallel and the other with the light globes in series.

(3 marks)

Description		Marks
Parallel	Series	
		
Cell correctly inserted in circuit		1
globes in parallel		1
globes in series		1
Total		3

- (b) Assuming that all components are the same in each circuit, explain which circuit will have the brighter light globes.

(3 marks)

Description	Marks
Parallel is brighter	1
There is a higher potential difference across each globe OR lower total resistance	1
So there is more current through each globe	1
Answers may correctly address power consumption, eg. parallel is dimmer if the power source has a limited power output.	
Total	3

Question 9

(4 marks)

Select an article of the geologist's clothing and explain what property it has that would make it useful for the hot environment, and what form of heat transfer it is used to address.

Description	Marks
Article of clothing identified e.g. gloves	1
Method listed e.g. radiation	1
Property described e.g. shiny – reflects radiant energy	1–2
Total	4

Question 10

(4 marks)

- (a) Calculate the specific heat capacity (including the units) of the rock sample, given the rock absorbs 1.57×10^6 J of energy to increase its temperature by 1050°C . (3 marks)

Description	Marks
$c=Q/[m(\Delta T)]$ rearrange formula $=1.57 \times 10^6/[1.78 \times (1050)]$ values in right spot OR substitute values into equation	1
$= 840$ magnitude	1
$\text{J kg}^{-1} \text{K}^{-1}$ or $\text{J kg}^{-1} ^\circ\text{C}^{-1}$ units	1
Total	3

- (b) The geologist could have estimated the specific heat capacity of the rock by measuring the temperature of the water in the bucket before and after adding the molten rock and hammer to it. Give a reason why this method would not give an accurate result. (1 mark)

Description	Marks
Any valid reason, such as: - splashing/water loss, - production of steam - heat lost in transfer or to hammer	1
Total	1

Question 11

(3 marks)

Use the stick figure below to draw a free body diagram which clearly shows the forces acting on skater C.

Description	Marks
One or more horizontal forces e.g, pulling forces from other skaters 	1
one or more vertical forces e.g. normal  and gravitational 	1
Zero resultant forces (e.g. magnitudes cancel)	1
Total	3

Question 12

(4 marks)

Complete the following table by using the diagram(s) and your understanding of the concepts of thermal expansion and contraction to choose the best answer. Tick the appropriate column to indicate your answer to each question.

Question	Metal 1	Metal 2	neither
Which metal expands more when heated?	✓		
Which metal has particles with the greater average kinetic energy when in the “off” position?			✓
Which metal’s particles have the least change in their average distance apart when cooled?		✓	
Which metal’s particles have the least change in their average distance apart when heated?		✓	

Description	Marks
Metal 1	1
Neither	1
Metal 2	1
Metal 2	1
Total	4

Question 13

(4 marks)

A 0.250 kg ball bounces on a hard surface after being dropped from a height. The ball retains 80% of its kinetic energy in the collision and rises to a maximum height of 0.870 m above the ground. Calculate its potential energy at A.

Description	Marks
At B: $E_p = mgh = 0.25 \times 9.8 \times 0.870$	1
$= 2.13 \text{ J}$	1
$E_{pA} \times 0.8 = 2.13 \text{ J}$	1
$E_{pA} = 2.66 \text{ J}$	1
Total	4

Question 14

(3 marks)

Annie carries a suitcase into a lift. As the lift starts to move, she feels the suitcase change weight and feel lighter. Explain how she can tell which way the lift is moving.

Description	Marks
When the lift starts to move, it undergoes acceleration.	1
She is feeling the resultant force (weight) due to the combination of the acceleration of the lift and gravitational acceleration acting on the mass	1
For the bag to feel lighter, the lift needs to be accelerating in the direction of the weight force (down).	1
Total	3

Question 15

(4 marks)

(a) Explain what is meant by the term 'non-ohmic'.

(2 marks)

Description	Marks
Does not follow Ohm's law	1
A non-ohmic conductor does not proportionally increase the current compared to potential difference (resistance is not constant)	1
Total	2

(b) Calculate the resistance when the potential difference is 1.0 V.

(2 marks)

Description	Marks
$R = V/I = 1.0/2.5$	1
$R_1 = 0.40 \Omega$	1
Total	2

Question 16

(4 marks)

A plastic pen, when rubbed with wool, is then seen to be able to attract small pieces of paper from a desktop. Once the paper touches the pen, the paper flies off. Explain these phenomena.

Description	Marks
Rubbing the plastic creates a charge imbalance (adds/takes away electrons)	1
The charged pen attracts the paper	1
Upon touching the pen, the potential/charge between pen and paper equalises	1
Potentials/charges are now the same, so repel	1
Total	4

Question 17

(5 marks)

The following graph shows the temperature change when 1.00 kg of water is heated in an insulated container by a 2.00 kW heater. The temperature change has not been recorded.

Using the information in the graph,

- (a) determine the time interval, in seconds, between B and C; and (2 marks)

Description	Marks
Time taken from graph $20 - 1 = 19$ minutes (19 ± 1)	1
$19 \times 60 = 1140$ s (1080–1200)	1
Total	2

- (b) use this time value to calculate the amount of energy supplied by the 2.00 kW heater in the time interval between B and C. Express your answer with appropriate units.

(3 marks)

Description	Marks
$W = P \times t$	1
$= 2000 \times 1140$ $= 2\,280\,000$ J or 2 280 kJ or 2.28 MJ (includes 1 mark for the unit)	1–2
Total	3

End of Section One

Section Two: Problem-solving

50% (85 Marks)

Question 18

(16 marks)

- (a) The unit mAh is a common unit referring to the amount of stored charge that is available, where 1mAh (one milliamp hour) is equal to 3.6 coulombs. Show by appropriate calculation(s), that 1mAh = 3.6 C (3 marks)

Description	Marks
1A = 1000 milliamps	1
1 hour = 60 × 60 = 3600 s	1
$I = q/t$; $q = I \times t = 1/1000 \times 3600 = 3.6 \text{ C}$	1
Total	3

- (b) Fill in the first two columns of the table for the three power sources using information from the photographs. Then calculate the stored charge and enter the value in the third column. The first row has been completed for you. (4 marks)

Photograph	Potential Difference (V)	mAh	Stored Charge (C)
A	1.2	2450	8820
B	3.7	1050	3780
C	3.7	760	2736

Description	Marks
copying values from photographs	1–2
1 mark for each calculation	1–2
Total	4

- (c) Calculate the work that can be done by the AA rechargeable cell in Photograph A, assuming all of the charge could be released at the given potential difference of 1.2 V. (2 marks)

Description	Marks
$W = qV = 8820 \times 1.2$	1
$= 10\,584 \text{ J (10.6 kJ)}$	1
Total	2

- (d) Determine whether the following statements are true or false. (3 marks)

Description	Marks
F, F, F	1–3
Total	3

- (e) The cell in Photograph A completely discharges in one hour (3600 s) when connected to a circuit. Calculate the resistance of the circuit. Assume the potential difference of the cell remains constant during this time. (4 marks)

Description	Marks
$I = q/t = 8820/3600$	1
$= 2.45 \text{ A}$	1
$R = V/I = 1.2/2.45$	1
$= 0.490 \Omega$	1
Total	4

Question 19

(13 marks)

Jake is working on the top of a building which is 21.5 m above the ground. He drops both a nut and bolt.

- (a) The nut has a mass of $5.55 \times 10^{-2} \text{ kg}$ and the bolt has a mass of $2.51 \times 10^{-1} \text{ kg}$. Which of the two will reach the ground first? Explain your reasoning. (2 marks)

Description	Marks
Reasoning only e.g. one will be more resistant to moving through the air, so will fall slower (0 if only one goes slower without reasoning); or both items have negligible wind resistance, so both items will fall at the same rate because both accelerate at 'g'.	1–2
Total	2

- (b) Assuming the bolt falls straight to the ground and there is no wind resistance, calculate

- (i) the time it takes for the bolt to reach the ground; and (3 marks)

Description	Marks
$S = ut + \frac{1}{2}at^2$; $21.5 = 0 + 4.9 t^2$	1
$t^2 = 4.387755\dots$	1
$t = 2.09 \text{ s}$	1
Total	3

- (ii) the final velocity of the bolt as it hits the ground. (3 marks)

Description	Marks
$v^2 = u^2 + 2as = 0^2 + 2 \times 9.8 \times 21.8$	1
$v^2 = 427.28$	1
$v = 20.5 \text{ m s}^{-1}$	1
Total	3

- (c) The ground consists of soft soil and the bolt enters the soil and stops 2.25×10^{-2} m below the surface. Calculate the magnitude of the force that the ground has exerted on the bolt. If you were unable to determine an impact velocity in Part (b), use the value 25.0 m s^{-1} . (5 marks)

Description	Marks
$v^2 = u^2 + 2as$ OR $W = \frac{1}{2}mv^2$ $a = (v^2 - u^2)/2as$	1
$a = (0^2 - 20.5^2)/(2 \times 2.25 \times 10^{-2})$	1
$= -9339 \text{ m s}^{-2}$	1
$F = ma = 2.51 \times 10^{-1} \times 9339$	1
$= 2.34 \times 10^3 \text{ N}$ ($3.49 \times 10^3 \text{ N}$ if value of $v = 25$ used)	1
Total	5

Question 20

(14 marks)

- (a) In the circle on the diagram above, clearly write the letter 'A' for ammeter or 'V' for voltmeter, indicating which meter should be correctly placed here. Explain why you have made this choice of meter. (3 marks)

Description	Marks
A	1
Ammeter measures current	1
Needs to be in series	1
Total	3

- (b) R_2 has a resistance of 12.0Ω . Given the total resistance of resistors R_1 and R_2 is 3.00Ω , calculate the resistance of R_1 . (3 marks)

Description	Marks
$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$	1
$\frac{1}{3} = \frac{1}{R_1} + \frac{1}{12.0}$	1
$R_1 = 4.00 \Omega$	1
Total	3

- (c) The cell provides 1.50 V of potential difference to the circuit, which has a total resistance, R_T of 9.24 Ω .
- (i) On the diagram on page 22, use an arrow to indicate the direction of conventional current in this circuit.
Show through calculation that the current flowing through R_3 is 0.162 A. (3 marks)

Description	Mark
Clockwise	1
$I_T = V_T/R_T = 1.50/9.24$	1
Therefore $I_T = 0.162$ A is true	1
Total	3

- (ii) Calculate the current flowing through R_2 . (5 marks)

Description	Marks
$V_3 = I_T R_3 = 0.162 \times 6.24$	1
$V_3 = 1.01$ V	1
$V_{12} = V_T - V_3 = 1.50 - 1.01 = 0.489$ V	1
$I_2 = V_{12}/R_2 = 0.489/12.0$	1
$= 0.0408$ A (0.0406 if unrounded I_T used)	1
Total	5

Question 21

(15 marks)

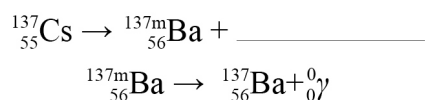
- (a) Determine the values for X and the number of neutrons Y

$X = \underline{\quad 37 \quad}$ $Y = \underline{\quad 3 \quad}$ (2 marks)

Description	Marks
$X = 37$	1
$Y = 3$	1
Total	2

- (b) Caesium-137 undergoes radioactive decay to form a short lived metastable barium-137m (m for metastable) and another particle. The barium-137m then undergoes gamma emission to become a stable barium isotope. (4 marks)

- (i) Complete the equation below for this decay chain, showing clearly the missing particle. (2 marks)



Description	Marks
Identifies beta decay	1
Properly shows ${}_{-1}^0\beta$ or ${}_{-1}^0\text{e}$ in equation	1
Total	2

- (ii) Outline **two (2)** properties that make caesium-137 a good choice for food irradiation. (2 marks)

Description	Marks
Produces gamma particles which has higher penetrating ability.	1
Reasonably long half-life	1
Total	2

- (c) A newly processed sample of caesium-137 ready to be used for irradiation would contain more of the unstable isotope _____ because _____

A 25 year old sample of Caesium-137 ready to be replaced would contain more of the unstable isotope _____ because _____

Description	Marks
Caesium-137	1
A new sample will contain mostly what was put in it	1
Caesium-137	1
Still has not yet reached its half life, so there should still be more than half left in the sample	1
Total	4

- (d) To increase the shelf life of some chicken meat, it is exposed to radiation. How much energy is absorbed by 1.50 kg of meat when given a dose of 3000 Gy? (3 marks)

Description	Marks
$E = \text{Absorbed dose} \times m$ (identifies absorbed dose)	1
$= 3000 \times 1.5$	1
$= 4.50 \times 10^3 \text{ J}$	1
Total	3

- (e) Name a precaution used by workers in this situation to limit their exposure to ionising radiation and explain how it reduces exposure. (2 marks)

Description	Marks
Precaution and explanation 1 mark each. For example: Use shielding – blocks radiation; Increase distance from source – decreases exposure; Use dosimeter – monitor exposure to limit; etc	1–2
Total	2

Question 22

(16 marks)

A 1.55 kg remote control car is accelerated from rest. The car takes 5.66 s to reach its maximum speed of 14.2 m s^{-1} .

- (a) (i) Calculate the acceleration of the car. (2 marks)

Description	Marks
$a = (v-u)/t = (14.2-0)/5.66$	1
$= 2.51 \text{ m s}^{-2}$	1
Total	2

- (ii) Determine the force needed to accelerate the car. (2 marks)

Description	Marks
$F = ma = 1.55 \times 2.51$	1
$= 3.89 \text{ N}$	1
Total	2

- (b) Calculate the gain in the car's kinetic energy. Express your answer with appropriate units. (3 marks)

Description	Marks
$E_k = \frac{1}{2} mv^2 = 0.5 \times 1.55 \times 14.2^2$	1
$= 156$	1
J	1
Total	3

- (c) Assuming no energy losses, calculate the power of the motor when travelling at its maximum speed. Express your answer with appropriate units. (3 marks)

Description	Marks
$P = Fv = 3.89 \times 14.2$	1
$= 55.2$	1
W	1
Total	3

NOTE: allow 1 mark for $P = \frac{\frac{1}{2}mv^2}{t} = 27.6$

- (d) Despite the force due to the power output of the motor, the car is unable to go any faster. Explain why the remote control car has a top speed when considering its motion in real life. Use one of Newton's Laws to help your explanation. (3 marks)

Description	Marks
Names law; states forces are balanced; concludes no acceleration. For example: 1 st Law; tend to stay in straight line uniform motion unless acted on by an unbalanced force; friction balances motor; or 2 nd Law; $F_{\text{net}} = 0$ because friction force is equal and opposite to the motor's force; so $a_{\text{net}}=0$.	1–3
Total	3

- (e) The front of the remote control car has a soft, bendable bumper that is attached to the car with a spring. Apply your understanding of change of momentum to explain how the bumper keeps the car from becoming too badly damaged in a collision. (3 marks)

Description	Marks
Momentum change is constant ($mv - mu = \text{constant}$)	1
The bumper increases the time taken during a collision	1
F goes down when t increases ($Ft = \text{constant}$)	1
Total	3

Question 23

(11 marks)

- (a) Calculate the specific heat capacity of the unknown metal given the above information. (4 marks)

Description	Marks
Setting them equal: $[m \times c \times (T_f - T_i)]_{\text{metal}} = [m \times c \times (T_f - T_i)]_{\text{copper}} + [m \times c \times (T_f - T_i)]_{\text{water}}$	1
Temperature change values in the right spots	1
$0.0787 \times c \times (95.0 - 25.0) = 0.0443 \times 440 \times (25.0 - 18.0) + 0.0750 \times 4180 \times (25.0 - 18.0)$	1
$5.509 \times c_{\text{metal}} = 136.4 + 2194.5$	
$c_{\text{metal}} = 423 \text{ J kg}^{-1} \text{ K}^{-1}$ (units not tested here)	1
Total	4

(b) It is difficult to determine the exact value of the specific heat due to laboratory conditions.

(i) What is one possible source of error that Jess and Ben might have encountered? (1 mark)

Description	Marks
Error source: Heat lost on transfer; to the surrounds; touching sample; specific measurement error, etc	1
Total	1

(ii) How would you expect the students' calculated specific heat to compare to the known value? (2 marks)

Description	Marks
Comparison of values acknowledges it is different (1 mark only) Greater (could be lower, depending on reason from (i))	1–2
Total	2

(iii) What could they do to improve their results? (1 mark)

Description	Marks
Improvement: Limit heat lost; repeat experiment; etc	1
Total	1

(c) Jess and Ben had to write a conclusion to their experiment, but found it difficult to distinguish between heat and temperature. Write a simple definition for the terms 'heat' and 'temperature' as they would apply to the experiment above. (3 marks)

Description	Marks
Heat – transfer of energy OR energy held by particles because of their position or movement	1
From hotter to cooler objects	1
Total	2

Description	Marks
Temperature – a measure of the average kinetic energy of the particles OR a measure of the hotness of an object	1
Total	1

End of Section Two

Section Three: Comprehension

10% (17 marks)

Question 24

- (a) The article states that 70.0 MW of thermal (heat) power are converted into 25.0 MW of electrical power. Explain why there is a difference in the quantities. (2 marks)

Description	Marks
The missing energy is waste heat	1
It is released into the environment through the cooling towers or Energy conversions are not 100% efficient	1
Total	2

- (b) The last paragraph states that the nuclear reactor can be used to replace an existing diesel-fuelled generator. In the diagram on page 30, circle the part of the power station that would be the same as for a diesel-fuelled power station. (1 mark)

Description	Marks
Circle around electrical section, can include cooling system, but not reactor.	1
Total	1

- (c) Using the concept of neutron-induced fission, explain how the boron carbide is able to 'shut-down' the nuclear reactor. (2 marks)

Description	Marks
The boron carbide absorbs the neutrons	1
Therefore fewer neutrons available for fission events to occur	1
Total	2

- (d) (i) Choose one point from the table or the article on page 30 that illustrates an advantage of using the HPM instead of a Fukushima-type of nuclear power station and explain why you believe it to be an advantage. (2 marks)

Description	Marks
Advantage, e.g. underground location	1
Related explanation, e.g. improved environmental isolation	1
Total	2

- (ii) Although advances in nuclear reactors have made nuclear power safer and more easily managed, there are still some problems. Using information from the article, the table or your course work, choose one point that illustrates a problem of nuclear power stations and explain why you believe it to be a problem. (2 marks)

Description	Marks
Problem, e.g. storage or disposal of spent fuel	1
Related explanation, e.g. waste is both toxic and radioactive	1
Total	2

- (e) Most power stations are able to provide electrical power at a potential difference of 1.10×10^4 V to local areas through a substation. Given an HPM produces 25.0 MW of electricity for a local area at 1.10×10^4 V, calculate the amount of current that is available. (3 marks)

Description	Marks
$25.0 \text{ MW} = 25\,000\,000 \text{ W}$	1
$I = 25\,000\,000 / 11\,000$	1
$= 2.27 \times 10^3 \text{ A}$	1
Total	3

- (f) Over its 10 year operational life, the plant will convert the binding energy of nuclides through fission processes to produce a continuous output of 70.0 MW of thermal power.

- (i) Calculate how many joules of energy the reactor would deliver over the 10 year period. (2 marks)

Description	Marks
$E = P \times t = 70\,000\,000 \times 10 \times 365 \times 24 \times 60 \times 60$	1
$= 2.21 \times 10^{16} \text{ J}$	1
Total	2

- (ii) Using the calculated energy in (i) and the formula $E = mc^2$, determine the decrease in the mass of the nuclear reactor during this time. If you were unable to determine a value for energy in (i) use $2.00 \times 10^{17} \text{ J}$. (3 marks)

Description		Marks
$E = mc^2 = 2.21 \times 10^{16}$	$E = mc^2 = 2 \times 10^{17}$	1
$m = 2.21 \times 10^{16}/(3 \times 10^8)^2$	$m = 2 \times 10^{17}/(3 \times 10^8)^2$	1
$m = 0.245 \text{ kg}$	$m = 2.22 \text{ kg}$	1
Total		3

End of questions